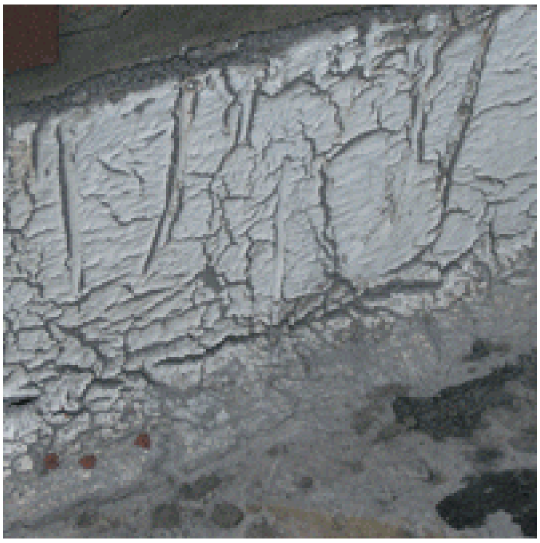
At a glance

In the late 1960s, aside from metal coverings, the choices facing a designer of a flat roof were generally fairly limited: either asphalt or built-up felt systems. Most roofs were of cold construction with minimal insulation while solar control was generally provided by chippings or promenade tiles.

Since then, much has changed. While warm roof systems are now almost universal, there are a bewildering number of different membrane options, all of which, if properly detailed, can deliver a good life expectancy. Defects in roof coverings are often due to inappropriate selection and use rather than a failure of the material itself (save for the expiration of normal life expectancy) with detailing accounting for a high proportion of failures. This subsection deals with the various types of covering in common use together with some of the defects to look out for. Common themes of workmanship and sensible design run throughout the various roof types.

Broadly there are 2 variations of flat roof:

- cold roofs (in which the insulation is located at ceiling level); and
- warm roofs (in which the insulation is located either immediately below the roof membrane or above it (inverted roof).



Earlier flat roofs tended to be of cold construction and were prone to condensation problems, although roof membranes such as asphalt performed well because the roof deck could act as a heat sink. With the gradual introduction of greater amounts of roof insulation, warm roofs became more popular. However, when insulation was located immediately below the membrane, the greater surface temperatures that developed would lead to quicker deterioration of the membrane. To avoid these problems inverted roofs became more popular. See Different types of roof construction .

Asphalt roofing performs well when it is protected against extremes of temperature. The material is thermoplastic and when heated tends to flow under gravity if unconstrained. At cold temperatures it becomes brittle and can be damaged under impact loads. Ponding on asphalt has not been shown to cause major problems but proper solar control measures are usually essential. See Mastic asphalt .

Built-up felt roofs (BUR) performed badly at one time, but the advent of polyester reinforcement brought a sea change in the industry with much greater flexibility and fatigue resistance than before. Attention to good detailing is needed to prevent damage to the covering from deck or substrate movement. Blistering due to entrapped air can be a difficulty but unless water-filled or greater than about 600mm, blisters are best left untouched.

Single ply membranes have become more popular and EPDM, TPO and PVC systems are common. EPDM needs adhesive bonded seams while TPO and PVC can be heat welded. Lap joints in these systems are all vulnerable if poorly executed. Loss of plasticiser can be a problem with some types, leading to embrittlement and cracking over time. See Thermoplastic polyolefin roofing .

Liquid membranes are also common. Hot applied systems are often used for new work but need solar protection. Cold applied systems often incorporate a polyester fleece for reinforcement and can be very effective, particularly where there is difficult detailing. However, adhesion failure arising from contamination of the substrate is a problem and the systems should never be used as a substitute for poor design and detailing. When used carefully and with compatible substrates, the use of liquid membranes for remedial work is often a good alternative to traditional roofing systems. See Liquid waterproofing systems .

In the 1900s the concrete flat roof and the system-built flat roof heralded a new outlook for the design and construction of housing. Unfortunately they left a poor impression on the buying public, as they failed repeatedly due to a lack of tolerance to thermal movement and the poor performance of early felts. See Defective flat roofs in the 1900s .

Different types of roof construction

Cold roof

Insulation located at ceiling level; void or roof slab is insulated from heat loss from below and is therefore at a colder temperature than the room during cold weather. With this form of construction there is a risk of condensation forming within the roof void or on the underside of the slab, so good cross-ventilation is essential. Vapour control layers at ceiling level are essential, but not 100% effective.

Quasi cold roof

Roof constructed as above, but without the insulation layer. Risk of condensation forming either on soffits of slab, interstitially or on reverse face of roof deck, i.e. the cold surface.

Warm roof

Insulation placed on top of the roof deck but beneath the waterproof membrane. Vapour control usually placed under the insulation. This form of roofing has a much reduced risk of condensation as the ceiling or roof void is at a similar temperature to the roof. However, the system prevents heat from infrared radiation from being dissipated.

Inverted roof

A roof in which the insulation is placed on top of the waterproof membrane. No vapour control is needed. Roof void and or deck are maintained at similar temperature to the room. The insulation protects the membrane from damage but must be anchored down. An inverted roof is also a warm roof.

System failures and detail failures

Accurate diagnosis of a defect is an essential prerequisite to specifying a suitable repair. Understanding whether or not the failure is due to system or detail will assist in the selection of a suitable and cost effective solution without necessarily overspecifying or adopting an over-optimistic approach.

A system failure is simply a failure in the roofing material that could occur at any point over the roof as a whole. For example, a defective roofing specification (unsuitable insulation, missing vapour barrier, unsound deck) could promote multiple failures. If one defective area is repaired there is every likelihood that other defects could occur elsewhere for the same cause. In the case of a system failure, replacement of the entire roof is probably the only sensible option.

On the other hand a detail failure occurs at a specific point or points and is related in some way to the detailing of the roof covering at that position. Repair would be possible, with no reason to doubt that the remaining areas would perform adequately.

Examples of system faults include:

Basic cause	**Materials at risk**	**Effects**
Problems due to moisture	Affects organic materials such as chipboard, plywood, strawboard, etc. Results in swelling, sagging, degeneration due to biological effects, rot. Moisture damage to HAC roof decks - risk of corrosion to reinforcement or alkaline hydrolysis.	Swelling can cause compressive stress in the deck, relieved by bowing. If the membrane is fully bonded, tensile failure may occur. Degeneration of the deck will result in loss of support and rupture of the roofing system.
Problems due to temperature	Can affect certain types of rigid foam, resulting in shrinkage or expansion. Expansion movements in large concrete roof decks resulting in bowing.	Places stresses on the membrane at board joint positions, rippling or rupture of membrane.
Problems due to a combination of moisture sensitive and temperature sensitive materials	Typically affecting woodwool slabs.	As above.
Problems due to poor design	All.	Inadequate design leading to deflection of the roof or risk of wind

		uplift; increased flexibility, risk of overloading, etc.

Examples of detail faults include:

Basic cause	**Materials and locations at risk**	**Effects**
Moisture	Localised faults in laps of BUR roofing. Missing or defective isolation joint at perimeter or upstand details - BUR and asphalt. Defective upstand detailing, BUR and asphalt. Lack of provision for movement at movement joints or changes in structure.	Localised water ingress, blistering. Repairs can be executed locally; total renewal not required.
Temperature	Slumping of asphalt at changes in level, upstands, etc. Cracking due to brittle failure at low temperatures	Localised water ingress, blistering. Repairs can be executed locally; total renewal not required.

Mastic asphalt

Although a traditional material and with a proven life expectancy of up to 60 years in some cases, asphalt roof finishes, particularly in new-build commercial buildings, are giving way to liquid membranes. It is now rare to find asphalt being specified as a primary membrane in new build.

When installed upon an uninsulated rigid concrete deck, asphalt performs very well (assuming that the detailing and workmanship are correct), but its long-term performance with insulated decks (warm roof construction) is less satisfactory. A properly installed mastic asphalt roof should have a life expectancy of at least 25 years, but if inadequate attention is paid to solar control, heat dissipation and maintenance it can be significantly less than this. The durability of asphalt is reflected not only in terms of high temperatures but also low temperatures, where the same insulation layer can prevent or at least significantly reduce the amount of heat taken up from the structure.

BRE Information Paper IP8/91, *Mastic asphalt for flat roofs: testing for quality assurance*, considers the durability of different grades of asphalt. Following testing of various samples, BRE concluded that major failures in asphalt were more likely to be the result of contraction and embrittlement at low temperatures. Thermal shock (rapid change from high to low temperatures) was found not to be a significant factor in the performance of the samples.

The material

Asphalt is essentially a formless blend of asphaltic cement with graded fine and coarse aggregates. The material is liquid when heated but cools to a dense, cohesive and voidless mass on cooling.

Asphaltic cement is a mixture of bitumen or a mixture of refined lake asphalt (a natural product) with asphaltite (a material allied to bitumen but with a higher softening point and greater strength and stability). Bitumen is either taken from natural sources or distilled from petroleum; it is a thick viscous liquid, mainly non-volatile. Without aggregates, asphalt does not perform very well, so it is usually combined with graded coarse and fine aggregates made from crushed limestone, siliceous, igneous or calcareous rock.

During the 1950s and 1960s asphalt roofing was common on flat uninsulated decks. The gradual increase in thermal insulation during the 1970s meant that, although until then it had performed well as a strong and durable material, asphalt's life was becoming much shorter. At that time warm roof construction was common; it was not unusual to specify asphalt on top of cork, polystyrene or rigid foam board with a layer of chippings or solar reflective paint as protection. While chippings of limestone, granite, feldspar, etc. offered a good level of protection initially, these materials could be subject to wind scour leaving larger areas of roof unprotected. Solar reflective paint systems were thought to be a better alternative, but they lose effectiveness fairly quickly and offer little benefit to flat horizontal surfaces. (Solar reflective paint is however the usual choice for upstands.)

Asphalt is relatively inflexible at normal UK daytime temperatures but inevitably the asphalt roof surface is subject to solar radiation. In older construction some of this heat is dissipated away into the structure, but if the asphalt is placed on top of insulation, the heat cannot dissipate to the same degree. As a result the asphalt softens and is prone to elastic deformation. Slumping of upstands or the gradual sinking of plates or bearers into the surface of the asphalt are typical indicators of plastic deformation. Repeated cycles of cooling and heating lead to cracking, particularly if the asphalt is restrained in some way.

Solar reflective paint

Thermal cycling can cause problems with the solar reflective paint as well. The paint can apply significant stress to the surface of the asphalt leading to a pattern of regular cracking.

Various different methods of protection of roof surfaces have been attempted over the years, for example:

- silver reflective paint;
- white reflective paint;
- asbestos cement promenade tiles (and their non-asbestos counterparts);
- capping sheets;
- chippings; and
- inverted roof insulation systems.

The solar reflective paint applied to this upstand has shrunk upon ageing, resulting in severe surface cracking of the asphalt. In turn the cracked areas absorb more radiation, exacerbating the problem.

Until the late 1970s the predominant finish was chippings of one variety or another, often with asbestos-based promenade tiles ('Spartan' tiles) in walkways or balcony areas. Towards the end of the 1970s greater reliance was placed on solar reflective coatings. These gradually became more popular because they could be added to vertical or sloping areas. However, solar reflective paints were not without problems and if the incorrect materials were used their performance could be disastrous.

All roofs will absorb a measure of solar radiation, most of which is converted into heat energy. Since solar radiation contains a variety of different waveforms in the spectrum, some UV light can also be absorbed. UV light causes organic materials to degrade. Asphalt hardens following exposure to UV light, reducing its tensile strength and making it susceptible to movements in the deck or flashings, for example.

A highly reflective surface prevents or certainly reduces the amount of radiation absorbed, while a black or dark surface will absorb more. In terms of roof coatings, few (if any) materials prevent absorption of radiation, although they can mitigate the effects. Conversely, a coating can serve to reduce (or increase) the amount of heat radiated from a roof at long wavelengths, thus affecting surface temperatures at night. During a cold, clear night, heat is radiated to the night sky (see cold night sky condensation) resulting in a reduction in surface temperature to several degrees below ambient air temperature. This can render the asphalt brittle, while the change back to a warm temperature during the day can permit the formation of longitudinal cracking unless the effects are mitigated by a solar coating.

Asphalt hardens following exposure to UV light, rendering it less flexible. Expansion and contraction of the lead flashing to this abutment detail has resulted in tensile failure of the asphalt. Ideally, joints in the flashing should be welted rather than lapped, and the lead securely fixed to a suitable timber edge batten bolted to the structure. (*Defects in buildings*, HMSO, London, 1989)

Asphalt is thermoplastic, it softens on heating and exhibits creep characteristics that are not reversed upon cooling. Below a critical temperature (or glass transition temperature, Tg) the material changes from being relatively plastic to hard and brittle - and at greater risk of damage from impact loads. Solar reflective paint can extend the time during which the asphalt is below its transition temperature.

The glass transition temperature (Tg)

The glass transition temperature provides a useful means of comparing the cold weather characteristics of a variety of polymers.

The chains of molecules that form a polymer can be likened to a plate of spaghetti. When warm, the individual strands slide over one another smoothly; this is described as an amorphous state. As the spaghetti cools it gets stiffer, the strands are less likely to move easily and eventually reach a stage when no movement occurs at all. The temperature at which this occurs is called the glass transition temperature - at this point the spaghetti reaches its glassy state. The transition from amorphous to glassy state is critical in evaluating the performance of roofing materials.

The glass transition temperature can be lowered by introducing small molecules called plasticisers. Plasticisers increase the distance between the larger polymer molecules.

The beneficial effects of solar reflective paint can tail off quite quickly as a result of weathering, soiling or ponding on the roof. Initially, a high degree of visible light is reflected, but this gradually diminishes. Cleaning the surface in spring may restore some of the benefits, but regular re-treatment, perhaps every 5 years or so, is needed for adequate long-term protection.

Some paints are not tolerant of surface contamination or moisture; application to an unclean surface or to one that is damp following overnight dew will probably result in failure and large areas of paint flaking off within a short time. In some cases performance improves if the asphalt has been sanded prior to coating: this serves to give the paint a better mechanical key.

Research by BRE in the late 1970s sought to identify if there were any particular benefits in aluminium paints over white paints. Aluminium, which tended to be based upon a bitumen binder rather than a water emulsion, was a common finish but is less so now. BRE found that aluminium was successful in blocking infrared and UV radiation to about 65% when new, but that reflectance dropped off after a short period of time. The paint served to reduce emissivity but not by enough to compensate for absorption. BRE were unable to identify clear advantages of aluminium over white paints although they noted that in some cases, presumably as a result of low permeability, blistering was more prevalent with aluminium paint. (*Solar reflective paints*, Information Paper 26/81, BRE,Watford, 1981)

For a solar reflective paint to be effective, it must be compatible with the roof material and should be designed specifically for the purpose. Paints that are not specially formulated can cause more harm than good, especially as they shrink with age. The shrinkage can cause the surface of the asphalt to crack and in turn the cracked area absorbs more radiation than the surrounding area. The localised increase in temperature results in localised softening and expansion of the asphalt - an effect that is not reversed upon cooling. Gradually, the problem increases in severity, the cracking is extended and the asphalt deteriorates.

Blistering

Blistering of asphalt is a common problem, often due to:

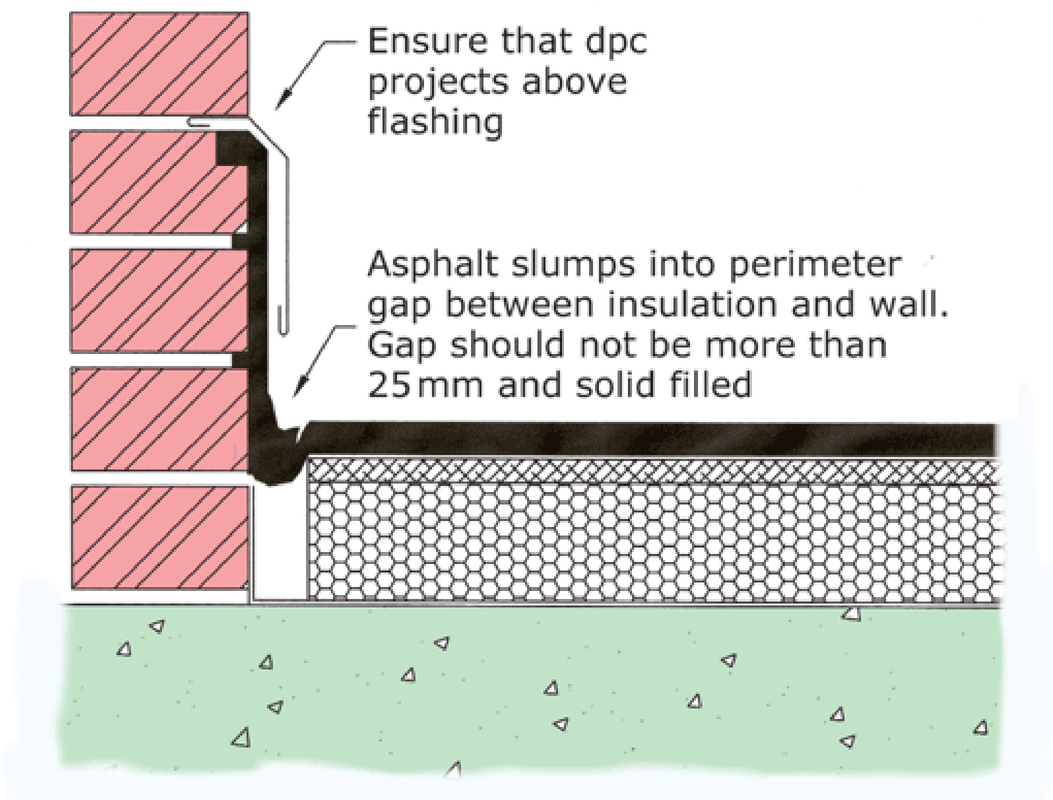
- poorly detailed upstands;
- a failure to provide effective vapour control; or
- water ingress into a roof deck.

The trapped moisture vaporises and forms a blister in the asphalt. Eventually, repeated cycles of vaporisation result in the perforation of a blister and further water ingress. See also text on blistering in Built-up felt roofing .

Upstands typically suffer from blistering as a result of poor detailing. Good construction practice dictates the use of raked joints in brickwork to provide a key to the asphalt and or the use of high bond primer to prevent the asphalt slumping. Ideally, the tuck at the top of a skirting should be protected by a lead or metal flashing. Frequently this is neglected, and a mortar joint substituted instead. Such a detail leaves the tuck vulnerable to water penetration. If a lead flashing is used, it is essential that the cavity tray is dressed over the top of the lead and not beneath it, otherwise water leaching from the cavity tray could be directed into the asphalt tuck and possibly behind the skirting. Similarly, the cavity tray must always project far enough to lap over the lead and not stop short of it. This problem is illustrated below.

Large vapour-filled blisters in an asphalt roof covering

The use of asphalt in warm roof situations has become far less prevalent and is best avoided. As an alternative, inverted roof construction has proved far more successful in maintaining the long-term performance of the roof membrane. The membrane is protected against mechanical damage as well as thermal cycling: being below the insulation, the asphalt is subjected to a much lower range of temperature variation.



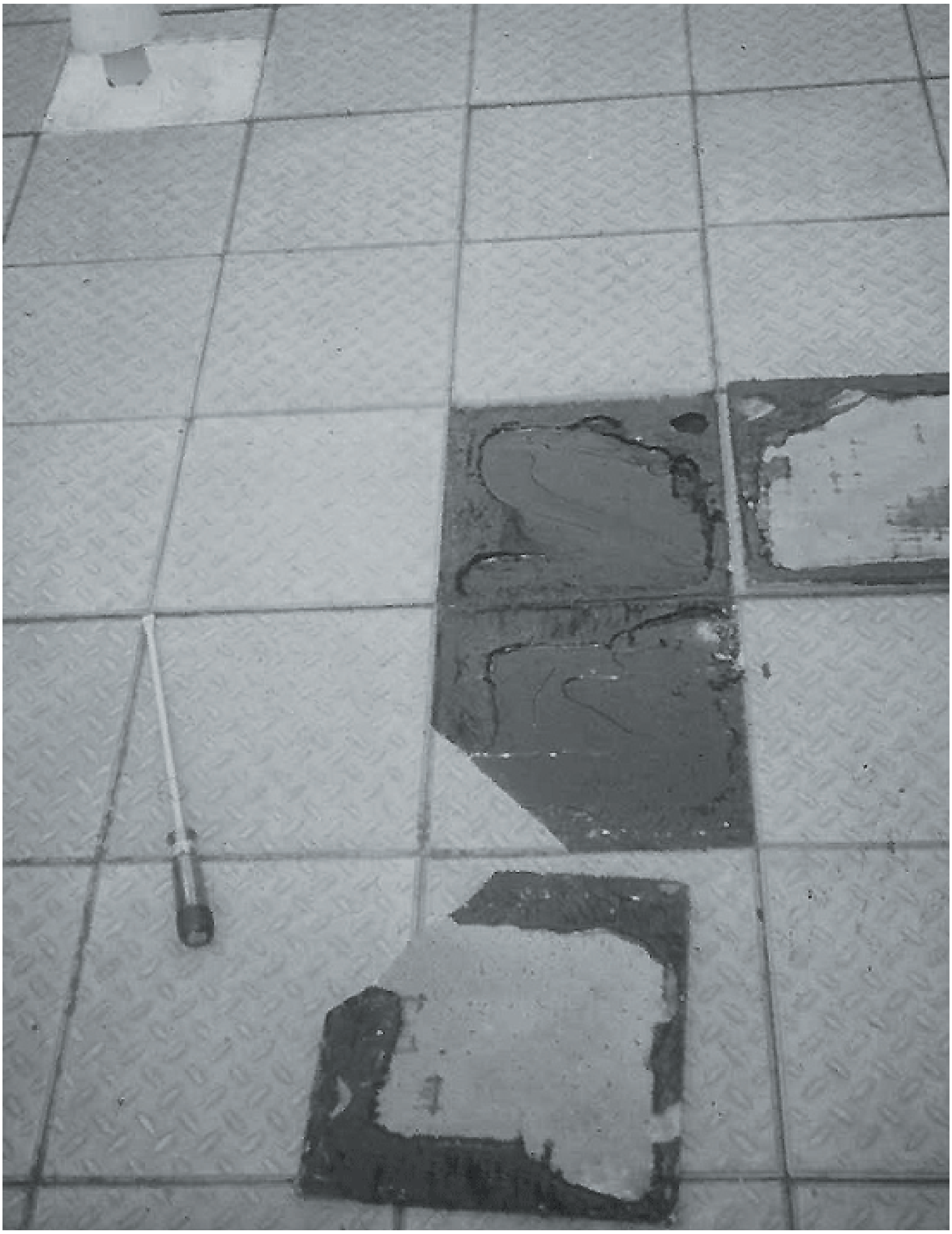
Gap between insulation and wall causes cold bridge and allows asphalt to slump.

Cavitation

Asbestos cement promenade tiles were a common finish to asphalt, offering good levels of solar as well as mechanical protection. However, smooth finish tiles tend to promote the growth of algae, in some conditions, making them very slippery and sometimes dangerous.

With the gradual restrictions on asbestos, cement fibre or GRC products became widely available. An early problem with these materials was their low permeability, meaning that water that became trapped beneath the tiles could not necessarily escape.

During the day, the surface temperature of the tiles could cause the trapped moisture to vaporise, a problem that was accompanied by the softening of the asphalt or bitumen bedding compound. Gradually, tiles could lift out of place, ultimately ratcheting upwards to present a trip hazard. However, while ratcheting was a problem, the formation and subsequent collapse of vapour bubbles gives rise to a problem of the gradual deterioration of the asphalt - in effect a process of cavitation. Cavitation is the formation of local cavities in a liquid, as a result of the reduction of pressure below a critical value. It frequently causes the pitting or wearing away of a solid surface.



Cavitation damage below impervious promenade tiles

To deal with the problem, promenade tiles were subsequently manufactured to be porous or vapour permeable.

Cracking

Asphalt is susceptible to deck movements, particularly following ageing and exposure to UV light. At low temperatures, the material is very brittle and can fracture under impact loads, or misguided attempts at opening up using a bolster. Similarly, embrittlement can result from overheating during laying.

Cracking is often found around junctions with alternative construction, perimeter details and projections as a result of differential movements in the background materials. If asphalt is laid on a lightweight or flexible deck (such as woodwool), movement between the deck and the enclosing walls must be accommodated by isolated upstands, otherwise failure occurs at the transition between horizontal and vertical work. Cracking at upstands can sometimes be caused not only by failure to provide for movement, but by poor workmanship during installation. For example, when the asphalter places the skirting, he/she will form a triangular fillet detail. If the asphalt is insufficiently cleaned before forming the fillet, poor adhesion permits cracking to form along its edge.

Another reason for fillet failure could be a lack of support at the edge of the insulation in warm roof construction. If a gap exists between the insulation and the enclosing wall, the asphalt can slump into the void over time, compromising the asphalt at the perimeter.

Deflection of the roof structure or collapse of a roofing screed due to poor compaction could create similar tensile crack patterns in the base of an upstand.

Examination of the crack patterns in asphalt will give some indications as to their likely cause. For example, cyclic movements in the deck arising from thermal or moisture changes are likely to produce compressive rippling type defects, while tensile cracking can occur as a result of the structural arrangement of the deck or background materials.

Asphalt skirting has been taken across a parapet movement joint without provision for movement: it has cracked as a result.

Ponding

Ideally a flat roof should never be totally flat. CIRIA (*Flat roofing: design and good practice*, CIRIA/BFRC, London, 1993) recommends a minimum fall of 1:80 or greater, although this is often not attained. Deflection of the roof deck can lead to the reversal of falls away from outlets and to ponding.

There is some debate as to whether or not ponding is a bad thing for asphalt roofs. The general consensus is that as far as asphalt is concerned, ponding is not necessarily harmful, although of course if a leak does occur there is a greater reservoir of water available to penetrate the building.

Rigid foam insulation

Since polystyrene (EPS) is a thermoplastic material it is easily damaged during the placing of asphalt. Therefore, its use as an insulant in conjunction with asphalt is generally limited unless suitably protected (at least one manufacturer supplies a felt faced material for this purpose). Furthermore, it can be dimensionally unstable over changes in temperature and this can lead to the opening and closing of the boards and problems of rucking or splitting in the waterproof layer.

Rigid foam boarding, usually polyisocyanurate (PIR), is a more popular choice because of its higher temperature resistance and good insulation properties. PIR is a thermosetting material, and rather than melting in the way EPS would, it tends to char. The material begins to degrade at temperatures above 150?C and so still requires protection in the form (usually) of bitumen-impregnated fibreboard.

Until the early 1990s almost all commercial rigid foam products used CFC-11 as a blowing agent. CFC-11 was held to be an ozone-depleting substance. Under various protocols it was phased out of use and substituted with, for example, HCFC 141b. The transition was thought to have led to various manufacturing problems and product failures, although it is probably more the case that poor mixing of the component parts was to blame.

PIR foams are generally stable, but roofing failures involving these materials are not unheard of and can lead to expensive remedial work, often involving re-covering. Typical defects are:

- formation of depressions in the roof after laying;
- shrinkage of boards creating board outlines in the asphalt;
- poor compressive strength; and
- cupping of boards giving rise to reflective undulations in the asphalt.

Most of the above problems can be attributed to manufacturing faults rather than issues of poor workmanship, save that improper storage of materials on site can give rise to performance difficulties.

The production of PIR and PUR foams involves the mixing of 2 component parts, part A and part B, to form a chemical reaction. Part A is usually Methylene Diphenyl di-Isocyanate (MDI). The catalyst, part B, is a mix of polyol, various additives and blowing agents. The mixed components then react exothermally (producing heat) to form a rigid thermosetting polymer. The maximum strength of the product is determined by the cell structure of the foam; spherical or slightly egg-shaped cells are needed and these are attained by mixing very precise quantities of A and B chemicals. If the ratio of chemicals is allowed to deviate during manufacture, there is a risk that the cell structure will be more elongated rather than spherical. Elongated cells are much weaker, with the result that the crushing resistance of the foam is less; it becomes brittle and is easily damaged.

Incorrect mixing ratios therefore account for the subsequent deterioration of the insulation, particularly in areas that are regularly trafficked during installation or afterwards. Poor compressive strength leads to crushing damage under foot, which may become manifest over a period of time after laying in the form of a series of depressions in the roof surface.

If severe enough, the deterioration of the boards results in the delamination of the fibreglass outer covering.

Edge cavitation

Edge cavitation of the boards is a defect where the edges gradually become concave, creating gaps and reducing thermal insulation. The problem is usually associated with poor strength as well. Again a manufacturing problem, the root cause appears to be the introduction of small quantities of water into the B chemicals to produce CO2 as a joint blowing agent. However, the water can dissolve some of the HCFC blowing agent resulting in an imbalance in the mixing ratio and subsequent weakness.

During mixing, the foam reaches temperatures of around 149?C. If CO2 is present it will diffuse out of the cells at a rate that is much faster than air at ambient temperature can replace it. This leads to an imbalance of pressure which, coupled with the weakened cells, creates an elongated cell structure. The problem occurs most rapidly at the board's edges, resulting in poor foam structure and board-edge cavitation.

Cupping and bowing of the boards can result either from manufacturing faults or from poor storage of the boards on site. The latter is easily explained if the boards are maintained in damp conditions prior to laying. Since they are usually supplied in bundles, the top or bottom sheet is at most risk if its outer surface is allowed to become saturated. If one face is wetter than the reverse face, differential expansion occurs causing the panel to cup. The problem is likely to be randomly spread over a roof and may be reflected in the asphalt layer.

Case study: Art Deco building, South coast

The building comprised a large L-shaped structure with an asphalt covered flat roof originally constructed in the 1930s. The roof had performed well in its time, but with no thermal insulation it gave rise to large heat losses. The designer of the re-roofing scheme concluded that the most appropriate form of repair would be to treat the existing asphalt as a vapour barrier and to provide thermal insulation in the form of polyisocyanurate boards, a layer of fibreboard and a new covering of polymer modified asphalt finished with solar reflective paint.

Within a few months of laying, depressions formed in the roof surface leading to ponding. The asphalt appeared to be performing well, but the freeholder expressed concerns over its long-term performance. The depressions were random in nature, but were more prevalent in areas of the roof where the asphalt boiler had been located - areas that would have received more foot traffic during installation.

A series of trial holes were cut into the asphalt. These revealed that while insulation was present it comprised 2 different types - a mixture of pink (PIR) and cream (PUR) types. Generally, the depressions occurred where the PUR had been used. In a few cases, moisture was found trapped within the roof construction.

Since PIR boards had been specified, the contractor had clearly used the incorrect material and was in breach of contract. However, theoretically there was no reason why the insulation should not have performed properly as the fibreboard layer should have been of sufficient thickness to have protected the insulation (which is thermosetting) from heat damage during the laying of the asphalt. Further core tests revealed a manufacturing problem with the boards leading to a lower than expected compressive strength. The cells of the board were elongated rather than spherical, which gave them a much weaker structure, easily damaged under load. Eventually the roof covering was replaced at the cost of the main contractor, the subcontractor having gone into receivership in the intervening period.

Alternatively, poor mixing, or poor mixing ratios, can lead to an uneven distribution of cells (in terms of size and shape), which create dimensional instability and different properties throughout the thickness of the board. When placed under load, the different properties react in different ways, with the result that cupping or bowing can occur. Such problems lead to the formation of regular ridges or depressions on the outline of the boards.

Case study: office building, west London

The building was concrete framed with in-situ concrete roof decks constructed in the early 1980s. At top floor level were some residential flats, each with a large balcony enclosed by concrete parapet walls. The roofs were of asphalt laid upon an expanded clay insulation system, 'Insulscreed' - small balls of expanded clay coated with bitumen. The insulation layer was around 150mm thick, with the roof draining towards a shallow gutter along its front edge.

To protect the asphalt, precast concrete pavings had been laid on plastic pedestals; there was no separate insulation or geotextile membrane. At the perimeter of the roof, a shingle margin had been provided.

One of the residents placed a series of wooden planters around the roof, planted with bamboo for screening purposes. After 5 years or so a small leak occurred in the roof and the roof was eventually exposed to reveal that the roots from the bamboo had broken through the lining to the planters and had entered the shingle margin. From here they had penetrated the asphalt and grown out again through the asphalt upstand.

Shoots of bamboo sprouting through the asphalt upstand

Bamboo roots had first penetrated the asphalt then spread through the expanded clay screed and exited out through skirtings. On examination, the roots had spread through the screed to affect nearly the whole roof. The black 'pebbles' visible above are the insulating screed, exposed after removal of the asphalt.

At first sight, the remedial work appeared simple: remove the planters, cut out the localised area of damage and make good the asphalt. However, on exposure it became evident that the insulating screed had created an ideal growing medium for the roots, being both warm and damp (from the leaks), and the roots had developed into a tangled mass that spread throughout the screed. Given the propensity of types of bamboo to propagate from roots, the risk was that even if the damaged areas were to be repaired, further plant growth could occur again within a period of months or years. The only safe solution would be to remove the roof and screed in its entirety and replace with new.

Case study: listed building, Kensington

The building comprised a terrace of 3 listed grade 2* large former houses in Kensington, converted for use as the headquarters for a large professional body. The building backed on to a typical London garden square. At basement and ground floor levels the building extended rearwards to form a large terrace at first floor level. The owner of the building used the terrace for corporate events and had for many years maintained a quarry tiled covering on its surface.

The terrace had fallen into disrepair and so it was decided to replace the asphalt covering with new and to provide a decorative finish of ceramic tiles. The specification included for the provision of foamed glass insulation cut to falls and laid over the original brick and filler joist vaulting. The roof was thus turned into a warm deck. The terrace extended over the rear of 3 houses, with a step at each party wall line. Asphalt had been dressed vertically up the steps to create a seamless membrane.

The tiles were laid using a flexible ceramic tile adhesive formulated for exterior use. Vertical tiles were placed at each upstand.

Within a few months of completion, tiles began to lift off and the asphalt began to slump at upstands and changes of level. Tiles covering the steps failed rapidly and became dangerous.

Eventually the problem became so bad that the entire finish was removed and replaced. The problems were due mainly to the tendency for plastic deformation within the asphalt. The layer of insulation beneath the membrane prevented heat that built up in the surface from dissipating. This resulted in higher working temperatures than would have been the case previously. Thus, the asphalt would soften and under gravity gradually tend to 'flow' over steps in the roof. Movement in the asphalt was greater than that which could be accommodated by the tile adhesive. Rigid grouted joints created compression stresses in the tile layer, which could only be relived by shear failure on the tile bed and 'popping' up of the tiles.

The difficulty for the designer of the remedial works was that in the intervening period the adhesive manufacturer withdrew its recommendation that the adhesives would be suitable for use on asphalt.

In addition, if ceramic tiles were refixed, the problem would reoccur.

The tiles were replaced with purpose made cement fibre tiles manufactured to resemble the original quarry tiles. Joints were not grouted but left open and compatible bedding adhesive was used. The solution was very much a compromise and while not matching the original in terms of quality did return the terrace to its original use.

Mastic asphalt performs well as a roof covering if applied properly and if the design of the roofing system is such that the effects of temperature are reduced as far as possible.

However, annual inspections are a sensible precaution. This is probably best carried out in the autumn so that leaves and other detritus can be cleared at the same time. In areas that are heavily affected by trees or other sources of debris more frequent inspections may be necessary. The early signs of failure can then be identified and corrective action taken.

The areas most likely at risk are abutment details, penetrations, movement joints, gutters and outlets. Blistering and slumping of asphalt, particularly to upstands, usually requires fairly urgent work to arrest deterioration and more water ingress into the substrate.

Retraction of asphalt around rainwater outlet details or edge flashings can also lead to problems of water ingress, and thermal expansion of flashings often leads to cracking at joint positions.

Built-up felt roofing (BUR)

At one time, the general rule of thumb was that a roof covered with felt to BS 747 would last about 15 years before re-covering was necessary. However, with the advent of high performance materials, life expectancies of 25 years or more can be routinely expected. Like most systems though, the success of a roof depends not only on workmanship, but the design of the roof in the first instance.

Early roofing felts employed an asbestos base, although these performed poorly and had poor tensile and tear resistance. In the 1970s, asbestos was replaced with polyester giving rise to a significant improvement in performance. For example, BRE fatigue tests on samples of BS 747 felt failed after around 32 cycles, while polyester samples failed after 3,200 cycles. By using polyester based felt and modified bitumen over 30,000 cycles were attained.

Contemporary construction has adopted the nomenclature of RBMs (reinforced bitumen membranes); guidance for the selection and use of these materials is given in BS 8747:2007. BS 747 was amended in 2000 but withdrawn in 2004 to be replaced by BS EN 13707:2004. Generally speaking the range of RBMs and their performance characteristics are now far superior to anything available in the early 1970s.

Whereas the norm for BUR roofs was at one time 3-layer felt, the tendency in current practice is to use 2 or more layers or alternatively a single-layer system using a specifically designed RBM. In a 2-layer system the lower sheet provides the flexibility and the ability to accommodate movements in the deck, while the top sheet provides a protective function against static or impact loads. Many top sheets are self-finished with either mineral chippings or foil to provide solar reflectance. These sheets are generally referred to as cap sheets.

To prevent the effects of oxidisation embrittlement, modern felts are usually compensated by the addition of either:

- **APP** (atactic-polypropylene): usually for torch-on felts. These plastomeric modified bitumen membranes are produced on polyester, glass fibre, and glass/polyester mixed bases. The base is saturated and/or coated with bitumen, modified with polyolefin or a polyolefin copolymer.
- **SBS** (styrene butadiene-styrene): usually pour and roll, bonded or torch-on applications. These elastomeric modified bitumen membranes are produced on polyester, glass fibre, and glass/polyester mixed bases. The base or carrier is saturated and/or coated with bitumen that has been modified with thermoplastic rubbers.

As with asphalt coverings, the success of an installation depends on the suitability of the deck to which it is laid and to the standard of detailing of abutments and roof penetrations. A deck that is likely to move and deflect under load or moisture changes is also likely to impart significant stresses to the covering, which offers less tensile strength as it hardens on ageing and exposure to UV light.

No matter how high the specification of RBM, it will be let down by poor attention to detail. Here, a membrane has been bonded over an older, asphalt membrane. The detailing of felt is almost impossible to achieve properly in these circumstances, with the result that attempts have been made to seal the top edge of the felt using an incompatible mastic sealant.

Problems with polystyrene insulation

Polystyrene insulation is generally considered to be unsuitable for asphalt roofing. Equally it can cause problems with BS 747 type felts because of their poor flexural strength. Boards should preferably be adhesively bonded to the deck rather than mechanically fixed; they must be tightly butted and secured to prevent wind uplift. Some insulation boards are intended for use with BUR roofing systems and sheet sizes are limited. Large sheets exhibit correspondingly greater movement capabilities at joints.

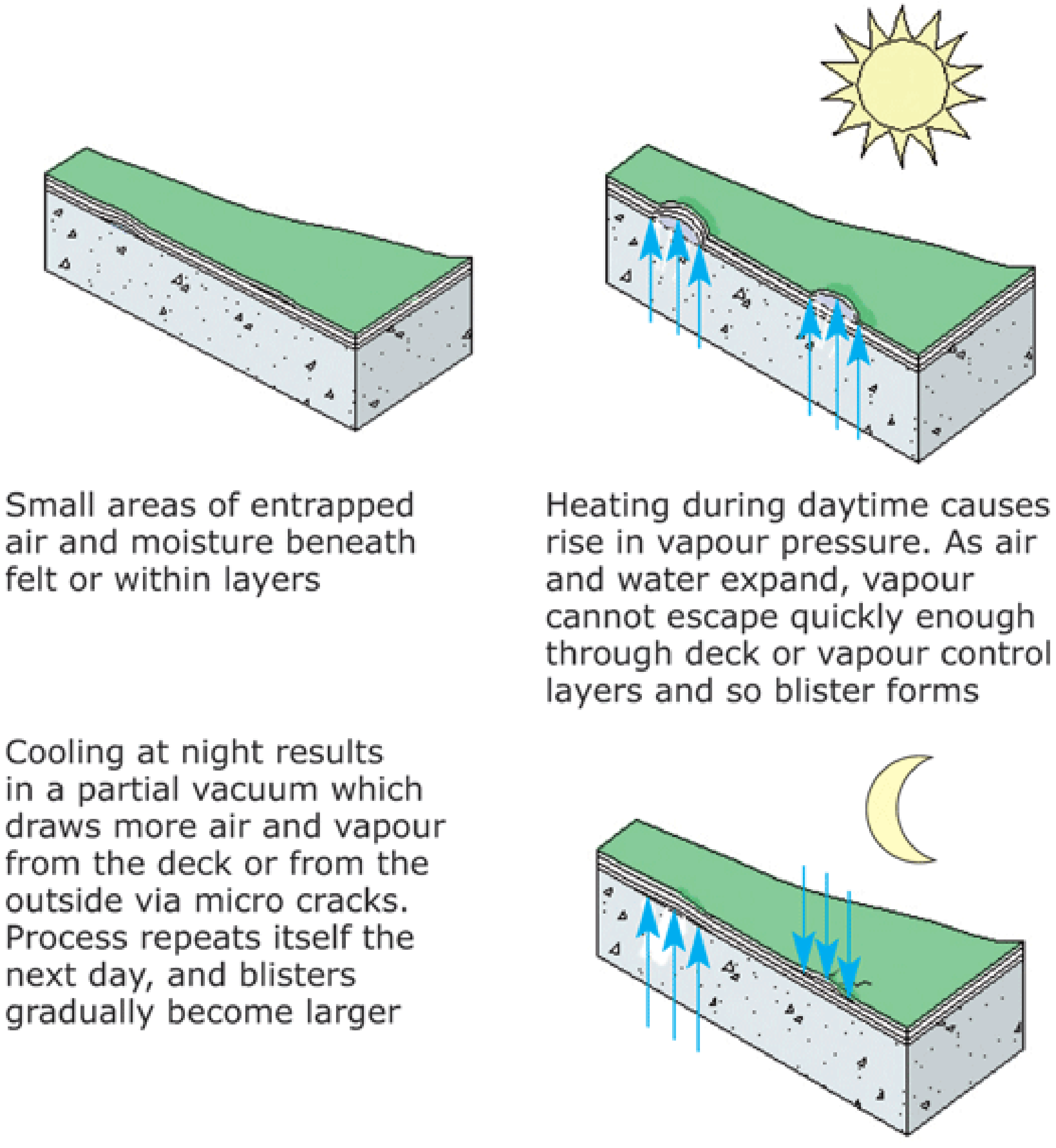
Felt used as a patch over existing failed asphalt. This may be satisfactory as a short-term measure, but here trapped moisture has formed severe blisters. The felt is also cockled and split.

If the boards are mechanically fixed, their propensity to move is greater, and cyclical movements can stress the covering, possibly causing it to rupture. Similarly, changes in level at board joints brought about by inadequate or partial fixing can damage the roofing membrane. Modern high performance materials may be better able to withstand the amount of movement than older BS 747 felts.

Blistering

Cockling or blistering of felt roofing is usually a symptom of water entrapment within the interlayers or beneath the complete system. The problem can occur wherever there is a lack of adhesion between the cap sheet and the base layer. Blisters could be due to residual water from construction or other detail failures elsewhere on the roof. Similarly, application of a membrane over an already damp surface will cause problems, as will work in cold weather if the material has been laid in unfavourable weather conditions.

A blister is an entrapped pocket of air and moisture (sometimes water filled) that can range over a few square centimetres to several square metres in area. Research by IRC-NRC (Liu, K.K.Y., Paroli, R.M. and Smith, T.L., *Blistering in SBS Polymer Modified Bituminous Roofs*, Construction Technology Update No. 38, June 2000, National Research Council Canada) suggests that the problem of blistering can be more prevalent with SBS type felts than APP types. If the interface between sheets is voided, any moisture present within the void can vaporise during hot daytime temperatures. The volume change associated with the conversion of water to water vapour is in the region of 1,250 times greater, so even a minute quantity of moisture can exert considerable expansive forces on the felt. Given the felt will be warm and flexible, the expansion forces may be sufficient to overcome the bond strength of the membrane around the blister, tending to enlarge it.



Blistering in built-up felt roofing

At night the vapour condenses but since the void is now slightly larger, the felt is cooler and stiffer and may not return to its original shape. More air is drawn into the partial vacuum than was present at the start. The cycle is then ready to commence again.

Blisters do not generally affect the tensile strength of felt, but a rupture will make a large difference in tensile strength of the cap sheet. Small blisters can therefore be left alone so long as they are not in areas subject to foot traffic. Larger, particularly water-filled, blisters should be cut out and repaired. (Paroli, R.M. and Booth, R.J., *Ways to Reduce Blistering Built-Up Roofs*, Construction Technology Update No. 4, May 1997, National Research Council Canada.)

Woodwool cement boards

Woodwool slabs were very common in the 1960s and 1970s but are less so now, at least within the UK. Roof decks of (usually) pre-screeded and edge reinforced slabs are common in commercial buildings and schools - the latter being particularly popular when used in conjunction with 'Metsec' type lightweight steel beams.

Although a somewhat maligned material, woodwool could or can perform well if used in appropriate circumstances. In older buildings the felts used were generally unable to accommodate the cyclical changes in moisture and temperature that brought about dimensional strains. Commonly, woodwool slabs are believed to lose their integrity when wet, although this is by no means certain - indeed they have been used in external situations for sound attenuation purposes by the power industry. However, in common with many construction materials, consistently damp conditions are unsatisfactory and can lead to softening and warping of the slabs. Warping of the slabs produces stress in the felt covering, with the risk of splitting along board joints.

Removal of old coverings during remedial works can be problematic, particularly where the felt has been fully bonded to the screeded finish of the slab. In these circumstances the screed can be damaged easily, leading to delamination problems and disintegration of the slab, necessitating a deck replacement in addition to the roof covering.

A further problem in 1960s and early 1970s buildings was a general lack of vapour control. If the woodwool was used in effect as a warm roof, it would need a vapour barrier directly beneath it. Such a provision would be rare, with the result that the dew point would occur at some point within the thickness of the slab giving rise to a risk of interstitial condensation. In severe cases, the slab could warp.

One particularly vulnerable form of deck (now rarely found due to life expectancy) is Stramit compressed strawboard - an insulation board material made from straw sandwiched between layers of building paper. Stramit boards had to be laid in a direction corresponding to the grain of the individual straws; failure to do this could result in excessive deflection. Condensation occurring within the board could lead to rapid decay.

Deck failure (chipboard): the outline of roof joists is clearly visible

Fibreboard

Bitumen impregnated fibreboard is vulnerable to moisture movement and sagging over time, particularly when it gets wet. In contemporary construction it is unlikely to be used in isolation. However, it was a common form of roofing when used in conjunction with profiled metal decking supported by lattice beams - the fibreboard being laid over a felt vapour barrier on top of the decking and covered with 3-layer BS 747 felt. If condensation is allowed to form on the fibreboard it will sag between ribs, giving a characteristic multiple ridging of the covering above the profile ribs. Regular stripped pattern staining along the rib lines is a good visual indicator of this type of construction. (*Maintenance and renewal in Educational Buildings. Flat Roofs ? Criteria and methods of assessment, repair and replacement*, Design Note 46, Department of Education & Science Architects & Building Group, 1985.)

Effects of condensation on the chipboard deck in cold roof construction: the chipboard has lost its integrity

Ponding

Ponding of roofs of BUR construction can be problematic as water can gradually penetrate at lap joints and find its way into the roof structure. BRE identified ponding due to undersized roof joists as one of the main causes of failure in a survey of rehabilitation work on housing (*BRE* *building elements: roofs and roofing ? performance, diagnosis, maintenance, repair and the avoidance of defects*, BRE Report 302, revised 2000).

However, in a separate study into the performance of roofing on Crown Estate buildings, no evidence was found that the life expectancy of roofing was compromised by the existence of ponding (*Asphalt and built-up felt roofing*, BRE Digest 144). As with asphalt, the existence of ponding does mean that there is a larger reservoir of water available to enter the building should a leak occur.

Deflection due to undersizing (or bowing as a result of thermal expansion being constrained by the surrounding structure) may not be the only cause of ponding. Movement in decking materials (particularly organic materials such as plywood and chipboard) can cause more localised depressions and any roof that exhibits a regular pattern of such depressions must be regarded with caution until the integrity of the decking material has been established. A frequent problem in older roofs is condensation occurring on or within the decking material causing it to expand, sag or decay.

Solar control

BUR roofs benefit from solar protection in much the same way as asphalt. Mineral flakes or chippings are often incorporated in the cap sheet to provide solar control - very light or white finishes are possible. A common alternative to reduce reflectance is the use of mineral chippings bedded in hot bitumen. However, as with asphalt, these can be subject to wind scour and can provide a useful base for the formation of moss and algae.

Promenade tiles or a screed material called Paropa were sometimes used. Deterioration of the felt in lines between concrete tiles could occur due to their increased exposure and small cyclical temperature movements in the tiles.

Wind scour

Loose laid insulation in inverted roof systems must be secured in place with either loose ballast, paving slabs or by using composite sheets incorporating a cementitious finish. Some insulation slabs contain an edge interlock such that adjacent boards engage with each other and help to provide resistance to wind uplift. However, edge interlock by itself is unlikely to provide sufficient restraint, particularly when the size of the pressure waves travelling across a roof are at a similar dimension to the individual boards.

In certain circumstances, wind scour of ballast systems can lead to the migration of the ballast into heaps and even blow individual stones off a roof - a possible risk to passers by or adjacent glazing systems, vehicles and the like. Much depends on exposure and roof profile, topography and location of adjoining buildings. Small chippings, such as those used as a solar reflective coating, are unlikely to be particularly harmful, but larger stones have the propensity to cause serious damage. Further information can be found in BRE Digest 311, *Wind Scour of Gravel Ballast on Roofs*, 1986.

Understanding wind uplift

There are 2 main influences on a permeable covering to a roof:

- the global pressure field; and
- the local velocity field.

The global pressure field is the pressure that forms around the building or roof as a whole, influenced by topography, exposure, proximity of other buildings and of course wind strength.

The local velocity field is the zone of pressure around a subsidiary part of the building, such as the tiles on a pitched roof, insulation slabs, ballast. Essentially, a rough projecting surface (for example, individual tiles) disturbs the air flow and can make a large difference to the air pressure immediately adjoining the element. Conversely, a smooth finish permits uninterrupted airflow and so the local velocity field will be less significant.

The effects of either the local velocity field or the global pressure field depend on the size of the gaps between elements and the volume of air beneath it. So, as pressure fluctuations affect the external air, small gaps and a large void will mean a time lag before the pressure can equalise. During this time the pressure beneath the element may be higher than above it, causing wind uplift.

A larger gap and smaller void will achieve pressure equalisation much faster and so the effects of uplift will be reduced. In effect this is why felt or timber sarking is placed immediately below tiles of slated roof coverings - it reduces the volume of the void and enables rapid pressure equalisation.

Roofs with loose laid insulation are vulnerable to wind uplift by this mechanism and so it is necessary to provide additional ballast or loading sufficient to overcome the forces generated by wind uplift.

Thermoplastic polyolefin roofing

Single-ply membranes have been popular for several years for both warm and inverted roof construction. Membranes usually comprise a reinforcement layer of polyester or fibreglass

matting or fibres sandwiched between 2 layers of flexible polymer.

Broadly there are 2 generic types (aside from the bitumen-based materials): thermosets (TS) and thermosplastics (TP). Thermosets are not modified by heating and therefore use bonded joints. A typical example of a thermoset single-ply membrane would be EPDM (ethylene propylene diene monomer). Thermoplastic materials are modified by heat and can be welded. The family of thermoplastic sheets includes thermoplastic polyolefins (TPOs), now a particularly popular material. The terminology is confusing because 'olefins' is not specific, it simply describes any molecule containing carbon-carbon double bonds or 'alkenes'.

TPOs do not contain plasticisers, they exhibit similar UV resistance and heat resistance as EPDM but are weldable. Some forms are fairly difficult to install as they tend not to hold shape and do not relax quickly.

The durability and life expectancy of TPO roofing materials is very good, but in common with many roof systems the success of the system relies heavily on good workmanship and the selection of materials that are appropriate for the particular circumstances in which they will be used.

According to a study conducted by NRC and others, membranes reinforced with continuous polyester fibre scrim had significantly higher tensile breaking strength than those reinforced with random short glass fibre mat. (Liu, K.K.Y., Paroli, R.M. and Smith, T.L., *Blistering in SBS Polymer Modified Bituminous Roofs* , ConstructionTechnology Update No. 38, June 2000, National Research Council Canada.) In North America, experience with TPOs appears to have been mixed; the product was marketed as a cheaper alternative to PVC roofing, but its use has been marked by seam failures and age-related embrittlement. However, the products used in North America tend to be thinner than their European counterparts, where there is greater experience in use. (Russo, M., *TPO problems require a closer look*, Roof Consultants Institute, Raleigh, USA, 1 July 2001.)

Some TPOs are reported to fade over time, but the most likely failure mode would probably be in detailing or welded joints. Precise temperature control is needed to ensure proper fusion. Too cold and an improper joint will be formed: too hot and the material may burn through. Since large scale welds are made by machine, the most likely problem areas will be at the start and end of a run.

In very cold weather conditions, the material may become brittle and suffer tensile failure. However, for the most part a review of manufacturer's data should enable selection to be made based on the likely glass transition temperature (Tg).

PVC roofing

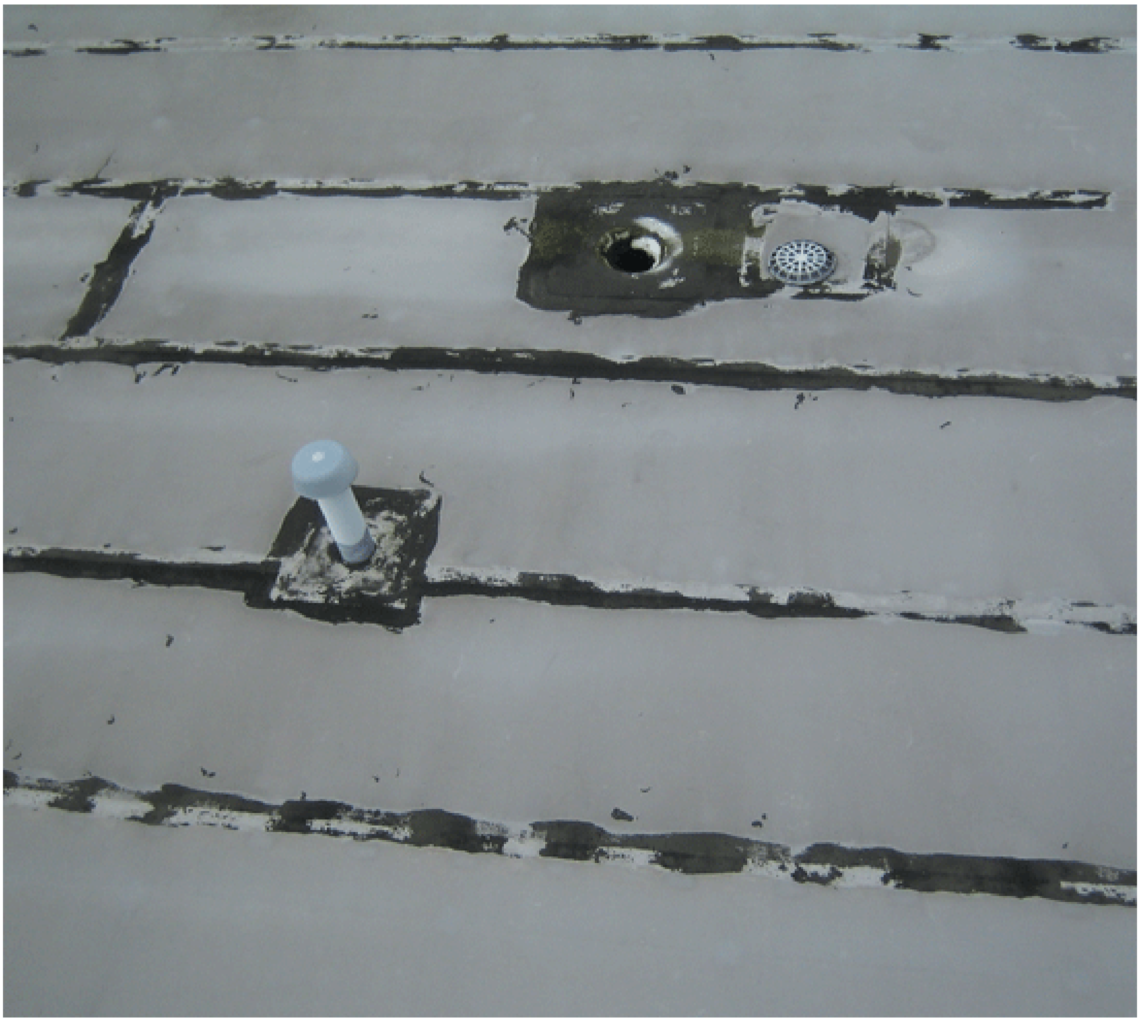
PVC roofing is another thermoplastic material capable of being welded in much the same way as TPOs. Performance of the material is generally good, with a design life of 30 years being expected. It was first developed in the early 1970s but suffered from age embrittlement and loss of plasticisers. Today, these performance difficulties have been largely overcome by the provision of polyester or glass reinforcement and plasticisers that are more stable.

Hot air and solvent welding is possible; early membranes suffered lap failures as a result of problems with solvent welding, and although this method is still used, heat welding using an electric hot air gun is more popular.

As with all sheet materials, the success of the roof depends on many factors, not just the durability of the membrane. PVC sheets can be fully adhered, partially attached, or loosely laid and ballasted. They are also available with a non-woven polyester fabric backing, which acts as a cushion to reduce the risk of perforation due to sharp ridges or material on the surface of the substrate.

PVC membranes are damaged by contact with bitumen or coal-tar pitch, as plasticiser from the PVC migrates into the bitumen, reducing the flexibility of the sheet and promoting premature failure. For this reason, an isolating membrane is usually specified.

Defects in PVC roofing are likely to follow similar patterns to some of the defects listed under EPDM , with seam failure and contamination issues being reported by ASTM Committee 8.



Attempts have been made to seal the joints in this PVC roof with an unsuitable sealant material

EPDM membranes

EPDM systems are often referred to as rubber roofs or membrane roofs. Other rubber and rubber-like materials used are chlorinated polyethylene (CPE), chlorosulfonated polyethylene (CSPE, trade-named Hypalon), and polychloroprene (Neoprene).

EPDM is a thermosetting elastomeric polymer that must be bonded using appropriate adhesives. The material is made from:

- clay, which gives fire resistance;
- carbon black to increase tear strength and resistance to ultra violet light;
- process oils for flexibility; and
- curing agents for vulcanisation.

Usually membranes are laminated together out of 2 plies. Like all single-ply membranes, workmanship and proper detailing are essential for proper performance.

EPDM roofs are not maintenance free and need to be inspected regularly to identify the early signs of system or detail failure. Manufacturers claim a life expectancy of at least 25 years for EPDM and despite early failures this does not appear to be optimistic.

Roof systems can be either fully bonded, mechanically fixed or loose laid and ballasted, the latter being unsatisfactory in conditions of high exposure because of the risk of wind scour. Performance data on EPDM roofing is not always easy to come by. Manufacturers are, for commercial reasons, disinclined to publish information on membrane failures or system faults. Continual product development does mean that troublesome details and methods are gradually improved and so it is not necessarily appropriate to judge performance on historic usage.

A 1990 study of EPDM membranes of between 6 and 13 years old revealed that a high proportion of roofs exhibited defects of one form or another. Of the sample roofs, 87% had had one warranty claim, while 67% had had multiple claims. (Wallace, T.J. and Rossiter, W.J., *ASTM Committee D-8 on Roofing, Waterproofing and Bituminous Materials*, ASTM International, 1990)

Many of the defects identified reflected workmanship and maintenance problems rather than inherent faults with the systems themselves. Judgment of the system based on old data would be unfair, since during the intervening period a number of advances have been made, such as improvements in bonding, elimination of neoprene flashings and their substitution with EPDM flashings, improvements to edge detailing and so on.

Typical defects with EPDM roof coverings, derived from ASTM Committee D-8 report, are summarised:

- **Puncture damage**: Most often caused by a failure to drive home the mechanical fixings to insulation boards fully, leaving the fixing pin slightly proud of the stress plate. The pin can then gradually work its way through the membrane under normal loading or foot traffic.
- **Embrittlement of flashings (depends on age of installation)**: Often neoprene based detail flashings at wall abutments or other perforations, exhibiting signs of checking, splitting and cracking where exposed to south and west and where overworked during installation - forming by hand or with roller tending to thin the material.
- **Contamination**: Contamination of flashings or field membranes with bitumen derived from previous roof coverings. Also contamination by grease or oil from roof mounted plant, extract equipment, etc. EPDM-based membranes are sensitive to hydrocarbons and should not be used in areas that are likely to become contaminated.
- **Delamination**: Interphasal delamination of insulation facing sheets when bonded to fully adhered EPDM membranes. Possibly due to manufacturing faults with the insulation (see rigid foam insulation). If this occurs, the roof membrane may be at risk from wind uplift.
- **Seam defects**: Birdsmouthing of seams, particularly at 3-way junctions due to poor workmanship, adhesion failure, contamination of contact surfaces, etc. Moisture working its way into seams can disrupt the integrity of neoprene-based splicing cement resulting in the gradual debonding of the entire seam. Since the 1980s, EPDM sheets have commonly employed factory applied adhesive seams rather than having to rely on site-applied adhesive.
- **Wrinkles in the membrane**: Possibly due to inadequate attachment, curling of insulation or contraction of the membrane after installation.
- **Poor dimensional stability of insulation**: Places undue levels of stress on the

membrane.
- **Defects in finishes**: Risk of puncture from sharp debris left in ballast, or wind uplift due to missing paving slabs, wind scour of ballast, etc.
- **Defects in membranes**: Due to poor or misguided attempts at repair using inappropriate or incompatible materials.

Liquid waterproofing systems

Numerous liquid applied roofing systems have been developed and specified for use in both new-build and repair situations. The performance of some coatings has not been satisfactory, but mainstream products are the result of considerable investment and research by producers and now offer considerable benefits over more traditional roofing systems or prefabricated single ply membranes.

Most of the new roofing materials are polymeric in nature meaning that they are made up of many ('poly') molecules ('mer') that are linked together in a chain; in effect forming very large molecules of different chemicals, for example, poly(propylene) or poly(isoprene). Liquid applied membranes offer certain advantages over the traditional membrane materials such as asphalt and built-up felt systems. They are lightweight and flexible and can be worked around complicated details that would otherwise be difficult to achieve. However, their convenience can lead to a risk of poor application or selection with the inevitable result that the roof performs unsatisfactorily.

In broad terms there are 2 types of membrane - those that are applied cold and those that are applied hot. Generally, the market for cold applied membranes lies in repair and maintenance - they are particularly suitable for application over pre-existing roof coverings without having the need to strip the old roof. Hot applied membranes are often used for new-build applications and require some form of surface protection.

Like all roofing materials, care in selection of the product and proper attention to detailing and surface preparation are key for ensuring long-term performance. Ease of application is no substitute for proper application.

European Technical Approvals

As a requirement of the Construction Products Directive, individual EU member states were required to prepare harmonised standards for liquid systems. This was achieved with the production of European Technical Approvals.

In the UK, a parallel system of certification is provided by the British Board of Agr?ment (BBA), w th many certificates now having been issued on a variety of liquid roofing products - some s ggesting an effective life of 25 years.

European Technical Approvals (ETAs) exist for:

- polymer modified bitumen emulsions and solutions;
- glass reinforced resilient unsaturated resins;
- flexible unsaturated polyester resins;
- hot applied polymer modified bitumen;
- polyurethanes;
- bitumen emulsions and solutions; and
- water dispersible polymers.

Solvent-based acrylic roofing systems are also available. These are not covered by ETAs and are often used for emergency repairs.

Type	**Formulation**	**Modifiers (to provide durability, flexibility, etc.)**	**Application**	**Roofing system**
Polymer modified bitumen emulsions and solutions	Aqueous emulsion or hydrocarbon solvent solution	Styrene butadiene styrene (SBS); styrene butadiene rubber (SBR); polybuylene (PB); rubber polyisoprene (IR); ethylene vinyl acetate (EVA); polychloroprene (CR); atactic polypropylene (APP)	Brush, spray or in some cases by spreader	Primer Emulsion or solution Internal reinforcement layer Solar reflective coating
Glass reinforced resilient unsaturated resins	Polyunsaturated resins with glass fibre (GRP)		Hand lay up process	Polyester resin basecoat/matting/ polyester resin second coat Chopped strand emulsion based glass fibre Polyester resin top coat Pigment and catalyst
Flexible unsaturated polyester resins	Unsaturated polyester resin reinforced with fleece		Spray or hand roller	Primer (type depends on substrate) Fleece Catalyst Pigmented unsaturated polyester resin
Hot applied polymer modified bitumen	Reinforced or unreinforced	Rubber, SBR, SBS or IR	Spread	May contain reinforcement worked in between coats, but must be protected and laid below 15 degree

				pitch
Polyurethanes	Many types, e.g. air or moisture curing, isocyanate component mixed with polyol or prepolymer		Spray or roller	Primer (depending on substrate) polyurethene base coat, fleece, polyurethene second coat and/or pigmented top coat depending on system and durability requirements. May include non-slip grit in top coat for walkway application.
Bitumen emulsions and solutions	Aqueous emulsion or hydrocarbon solvent solution	Clay or resin soap	Brush or spray	Multi-coat system Primer First coat Matting Second coat Solar reflective treatment
Water dispersible polymers	Many types, e.g. acrylic mastic, acrylic coating, co polymer, terpolymer, waterbourne rubber, SBR		Airless spray, roller or brush	Primer Reinforcement layer Polymer layer Non-slip grit

Most of the above products are cold-applied, but polymer-modified bitumens are very popular for new-build applications, while cold applied materials are generally (but not exclusively) used for remedial works. Both have good performance characteristics and if properly laid a life expectancy in excess of 30 years would not be an unreasonable expectation, although as with most materials this is highly subject to workmanship, protection, maintenance, etc.

Hot-applied rubberised bitumen membranes

By dispersing virgin or reclaimed rubber in asphalt together with limestone powder, oil and styrene butadiene-styrene (SBS) it is possible to manufacture a very flexible roofing membrane that can form a seamless membrane about 3-4.5mm thick. Some formulations include an additional spun bond polyester fabric interlayer with a further 3mm layer to form a roofing system. However, these materials are not usually resistant to UV light and so it is important to add a bonded membrane or alternatively a ballasted insulation layer to provide mechanical and solar protection.

The materials are usually supplied as cakes encased in polyethylene wrapping, which are melted in a heater to around 200?C. The melt is discharged from the heater into a suitable container and applied to the roof using long-handled, rubber-bladed squeegees.

Prior to application, the roof surface must be free from contaminants. Liquid membranes depend on a bond to the surface, so all traces of dust, grease and oil must be removed. Concrete surfaces are usually primed with a suitable asphaltic solution.

Cold-applied liquid membranes

These membranes are polymeric compositions with one or more ingredients, such as modified asphalt or coal tar pitch, or various resins or elastomers, such as polyurethane, dibromobutadiene, silicone or acrylic. The bitumen-based solutions are usually applied as emulsions, brushed or rolled in place.

Resin-based systems can often be supplied in different formulations to give 'guaranteed' life expectancies of up to 25 years. These materials can also be dressed with sand or mineral granules to provide slip resistance, and so are often used for waterproofing car-park decks, where a degree of friction is desirable and where various colour pigments can be used to advantage.

Cold-applied membrane showing excessive adhesion failure: probably laid over poor

felt roofing

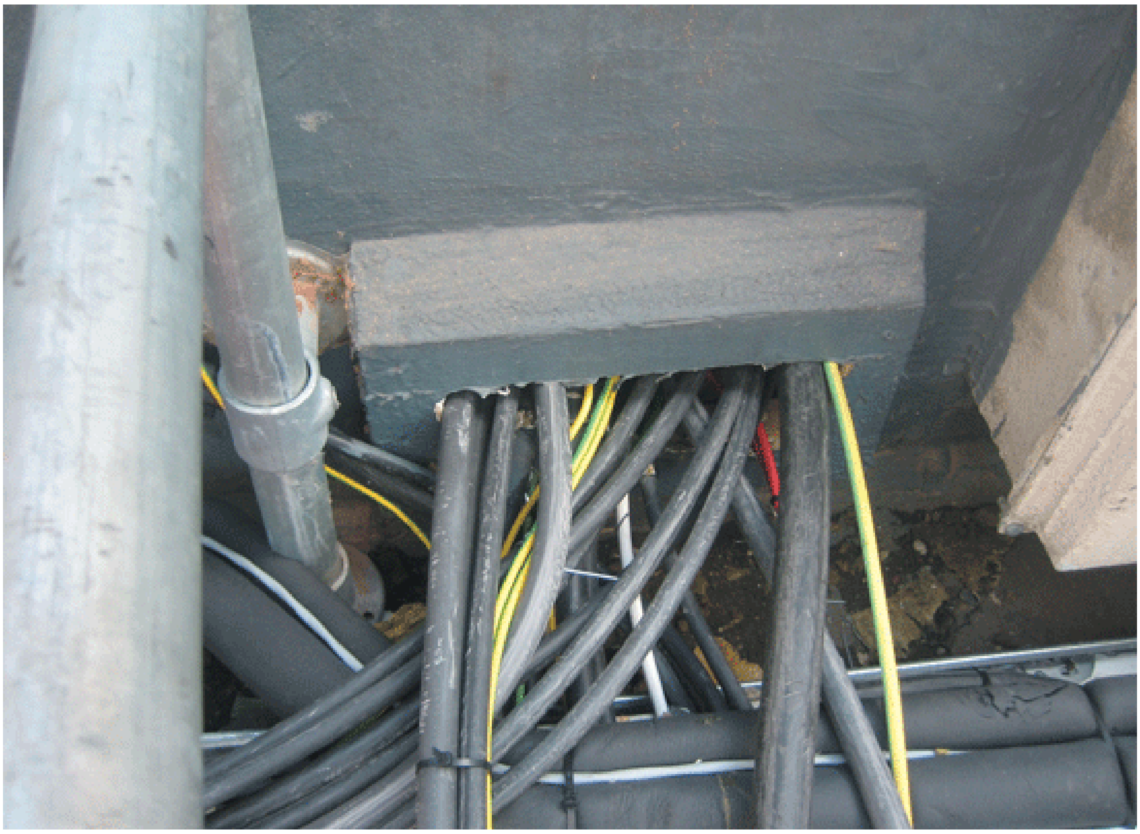
Cold-applied systems tend to be used for remedial operations, with the clear advantage that heat-producing appliances are not required and the membrane can be sprayed over existing finishes without the need for a temporary roof. The ability to form complicated weathering details with the minimum effort places the materials at a distinct advantage over BUR systems. Like hot-applied systems, it is usual to reinforce the membrane with a polyester fleece sandwiched between 2 layers of waterproofing. Some coatings require periodic refresher coats.

Limitations

While both hot and cold applied systems are relatively easy to lay, this does not mean that they tolerate lower levels of workmanship. Most systems require good substrate preparation - clear of sawdust, swarf, dirt, hydrocarbons or organic materials, while the maintenance of the correct dry-film thickness is important to ensuring future durability. Similarly, many products have only a limited shelf-life before they deteriorate (6 months or so).

From a designer's point of view, a failure to consider the usual principles of flat roofing design, rainwater disposal and the design of roof penetrations will lead to failure in much the same way as it would using more traditional methods. Similarly, because one of the great benefits of liquid membranes is their ability to bond to the substrate (and hence limit the propensity of water to travel through a sheathing layer) it is vital to ensure that the correct level of preparation is used and that the correct primers are employed. Compatibility with the substrate (and any likely contamination of the surface) must be established at the outset.

Incompatibility between substrate and coating has led to extensive peeling and delamination



One of the benefits of liquid roofing is its ability to accommodate difficult detailing, but it is no substitute for a failure to design details properly initially

Typical defects

- **Embrittlement of membrane due to loss of plasticiser. Regular pattern of fine cracks, material loses flexibility**: Possibly due to lack of UV protection (hot applied system) or contamination.
- **Tearing, ruckling or splitting**: Inability to deal with tensile stresses at places of concentrated movement, initiated, for example, by shrinkage in low temperature or creep during hot weather.
- **Splitting of membrane, lack of sufficient elasticity**: Ability to bridge cracks - particularly those that open and close cyclically as a result of changes in temperature or structural movements. If a membrane is fully bonded, the percentage elongation over, for example, a 3mm crack is very high.
- **Puncturing**: Debris or sharp materials dropped on roof, foot traffic, puncturing by plant growth.
- **Biological attack**: Colouring/mottling, softening and degradation.
- **Loss of adhesion**: Contamination of substrate with water, oil, sawdust, bitumen, etc. Incompatibility of background material.

Defective residential flat roofs in the 1900s

Flat roofs evolved from the influence of the modern movement and were consistent with the fashion for clean lines and abstract designs. The room layouts of these modern homes in the 1900s were in sharp contrast to the layouts of traditional homes, and the resulting footprints made it difficult to install pitched roofs.

The concrete flat roof and the system-built flat roof heralded a new outlook for the design and construction of housing, but were unfortunately to leave a poor impression on the buying public, as they failed time and time again due to a lack of tolerance to thermal movement and the poor performance of early felts. Flat roofs from this era are more than likely to have been upgraded or re-covered in modern materials, and while some may remain troublesome, most are performing well.

Roofs coated with asphalt outlasted many of their felt counterparts, and if recoated should be performing well. Concrete roofs are known to have had quality problems, and these usually need the early intervention of a structural engineer to access structural integrity. Condensation problems caused by cold bridging in concrete roofs can also be a problem, as no insulation was included on such roofs.

Common problems with flat roofs:

- **Roofing felts that bubble and split**: Patch repairs can be undertaken, but once an old domestic roof begins to fail it is unlikely to respond well to temporary repairs such as liquid sealants.
- **Leaks**: Mainly caused by membrane damage, which in turn soaks the decking and starts the decay process. If patching, ensure that it is a large patch, and if the problems are suspected to be the upstands then it is important to lift flashings or remove renders to see the detail close-up. Adding sealants on top of upstands has minimal success.
- **Blocked gutters and outlets**: These cause the roofs to fill and breach upstands. Wire balloons can be more of a problem than a cure on inaccessible high roofs. Sometimes it is easier to remove them and place access traps lower down the pipe run, so that any debris can be removed quickly at ground level. Outlets become detached from an internal downpipe. In poorly accessible situations there is little option but to use liquid sealants. This defect can be the cause of difficult-to-diagnose leaks, because where a pipe run has a horizontal section to it, water that leaks out of an outlet joint tracks on the outside of the pipe for some distance, before it forms a drip and falls. It is often an unfortunate feature of horizontal rainwater pipe runs that the reason it is horizontal is to carry it over an important area like a foyer or shop unit. The introduction of overflows and double downpipes can avoid the majority of problems associated with roofs becoming blocked.
- **Thermal movement**: Keeping a flat roof still is a good philosophy when assessing ways to keep a roof in good order. The use and maintenance of solar reflective paints and protective layers will stop stresses building up on roof coverings. The shadow cast by a taller adjacent building moving across a roof on a hot day can have a dramatic effect in terms of the temperature difference between dark and light, and can set up stress cycles which can tear open otherwise sound joints.
- **Flush eaves boards are prone to rotting** where the back face was not prepared sufficiently prior to fixing. Splice repairs of such items are not cost effective and complete replacement in many case is quicker. If it can be done easily it is best to re-detail these to stand proud of the wall on spacer blocks.
- **The roof is being used as a balcony**: This happens by degrees, where bedroom windows are converted to doors, or a roof designated as a fire escape is used by residents. These surfaces can be easily damaged by chairs and cigarette burns, and

should be preserved from such use by tenants. In private accommodation the use of lightweight tiles can help protect the surface. Items such as roof decking or plant pots should not interfere with the water run-off route to the outlet, as this can cause the roof to flood in severe weather and where there is slow water run-off.

- **Leaks in skylights**: These mainly arise due to poor design, where the cloaking over the upstand either does not allow for easy access to dress the felt up during a re-roof, or the cloaking does not cover the upstand sufficiently to protect the detail against wind-driven rain. It is unwise to assume that skylights 'lift off'. They may come off, but fixing them back can be difficult and the use of adhesives to fix frames is often relied on too heavily.
- **Staining to ceilings**: A lack of insulation in early roofs can make roof voids very cold, and with no vapour-check, condensation can occur within the roof space, which can cause damp staining. It is common to find that very simple constructions, that were originally only intended to be sun-rooms or potting areas, have been converted into a living space. Any attempt to improve the performance of such areas usually results in a major rebuild as the scope to add on all the necessary insulation, ventilation and vapour-checks is limited.